

Mean Value Theorem

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Name: _____ Date: _____ Score: _____

Determine whether the MVT hypothesis is satisfied; if so, find c. Show all work.

AB Calculus Worksheet #31

For #1 – 12, determine whether the hypothesis of the Mean Value Theorem or Rolle's Theorem is satisfied.

If it is, find c. If not, tell why not.

1. $f(x) = x^2 - 2x - 3$ on $[-1, 3]$	2. $f(x) = x^3 - x$ on $[0, 1]$
3. $f(x) = \frac{x^2 - x - 6}{x - 1}$ on $[-2, 3]$	4. $f(x) = \sin x$ on $[0, \pi]$
5. $f(x) = \tan x$ on $[0, \pi]$	6. $f(x) = x^2 - 2x + 1$ on $[1, 3]$
7. $f(x) = 4 - x^2$ on $[1, 4]$	8. $f(x) = x^2 - 7x$ on $[1, 5]$
9. $f(x) = 3x^2 - 5x + 1$ on $[2, 5]$	10. $f(x) = x^{\frac{3}{4}}$ on $[0, 16]$
11. $f(x) = \sqrt{1 - x^2}$ on $[-1, 1]$	12. $f(x) = \frac{1}{x^2}$ on $[-1, 1]$
13. $f(x) = \frac{1 - \cos x}{x}$ when $x \neq 0$ and $f(0) = k$. Find k so that f is continuous at $x = 0$.	
14. $f(x) = \frac{x^3 - 1}{x - 1}$ when $x \neq 0$, and $f(1) = k$. Find k so that f is continuous at $x = 1$.	
15. $f(x) = \begin{cases} \frac{x^3 + 8}{x + 2} & \text{when } x \neq -2 \\ k & \text{when } x = -2 \end{cases}$ Find k so that f is continuous at $x = -2$.	
16. $\int \sin^3 4x dx$	
17. $\int \sin^2 4x dx$	
18. $\int \frac{1}{2^x} dx$	



Mean Value Theorem

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Answer key — for instructor use only.

Answers:

1. $c = 1$	2. $c = \frac{\sqrt{3}}{3}$	3. Doesn't satisfy MVT because $f(x)$ is discontinuous at $x = 1$	4. $c = \frac{\pi}{2}$
5. Doesn't satisfy MVT because $f(x)$ is discontinuous at $x = \frac{\pi}{2}$	6. $c = 2$	7. $c = \frac{5}{2}$	8. $c = 3$
9. $c = \frac{7}{2}$	10. $c = \frac{81}{16}$	11. $c = 0$	12. Doesn't satisfy MVT because $f(x)$ is discontinuous at $x = 0$
13. $k = 0$	14. $k = 3$	15. $k = 12$	16. $-\frac{1}{4}\cos 4x + \frac{1}{12}\cos^3 4x + C$
17. $\frac{1}{2}x - \frac{1}{16}\sin 8x + C$	18. $\frac{-2^{-x}}{\ln 2} + C$		

